MILITARY SPECIFICATION

RELAYS, ELECTROMECHANICAL, GENERAL PURPOSE, NON-HERMETICALLY SEALED, PLASTIC ENCLOSURE (DUST COVER)

GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general requirements for general purpose relays that perform the switching functions associated with ground training systems and similar controlled environment condition applications.

CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

1.2 Military part number. The military part number shall consist of the letter "M" the basic number of the specification sheet and an assigned dash number (see 3.1) as shown in the following example:

M	83520/1	-001
Military designator	Specification number	Dash number

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

J-W-1177 - Wire Magnet, Electrical.
ZZ-R-765 - Rubber, Silicone.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-T-60 - Tape, Packaging, Waterproof.
PPP-T-76 - Tape, Packaging, Paper (For Carton Sealing).

MILITARY

MIL-I-10 - Insulating, Materials, Electrical, Ceramic, Class L.
MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-C-17 - Cables, Radio Frequency, Flexible and Semirigid, General Specification for.
MIL-P-116 - Preservation, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Air Force Acquisition Logistics Division, Electronic Support Division (AFALD/PTES), Gentile Air Force Station, Dayton, Ohio 45444, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MIL-STD-45662 - Calibration Systems Requirements.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern (see 6.2).

3.2 Qualification. Relays furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for the opening of bids (see 4.5 and 6.3).

3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the relays to meet the performance requirements of this specification. Materials used shall be self-extinguishing, and shall not support combustion, give off noxious gases in harmful quantities, give off gases in quantities sufficient to cause explosion of sealed enclosures, cause contamination of the contacts or other parts of the relay, or form current-carrying tracks when subjected to any of the tests specified herein. The selection of material shall be such as to provide maximum shelf life. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Metals. Metals shall be of a corrosion-resistant type or shall be plated or treated to resist corrosion, except that zinc, cadmium, or unfused pure tin plating shall not be used internally. Zinc or cadmium shall not be used externally. Unfused tin plating shall have a plating thickness of 200 microinches minimum if used for external parts.

3.3.1.1 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metal in contact, which tends toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy), is not acceptable. However, metal spraying or metal plating of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. Dissimilar metals shall be as defined in 6.4 through 6.4.4 inclusive.

3.3.2 Ceramic. Ceramic insulating material shall conform to MIL-I-10, grade L422 or higher. Ceramic used for external surfaces shall be glazed.

3.3.3 Magnet wire. Magnet wire shall conform to J-W-1177, except when wire not covered by J-W-1177 is required due to physical size or temperature range consideration; then a suitable wire may be substituted; as evidenced by meeting the performance requirements of this specification.

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...from the emergence of the lead from the relay).

3.3.4 Mercury. The use of mercury or mercury compounds is prohibited.

3.3.5 Magnesium. The use of magnesium or magnesium alloys is prohibited.

3.3.6 Rubber. Rubber shall conform to ZZ-R-765.

3.3.7 Fungus-resistant. Materials used in the construction of relays shall be fungus inert (see requirement 4 of MIL-STD-454).

3.4 Design and construction. Relays shall be of the design, construction, weight, and physical dimensions specified (see 3.1). Relays shall be so designed to insure proper operation when mounted as specified (see 3.1).

3.4.1 Enclosures. Relays shall be provided with a plastic enclosure to act as a dust cover to protect coils, contacts and other internal components from contamination.

3.4.2 Threaded parts. All threaded parts shall be in accordance with FED-STD-H28. Where practical, all threads shall be in conformity with the coarse-thread series. The fine-thread series may be used only for applications that show a definite advantage through their use. Where a special diameter-pitch combination is required, the thread shall be of American National Form and of any pitch between 16 and 36, which is used in the fine-thread series. Terminal threads shall be class 2A and 2B for external and internal threads, respectively.

3.4.2.1 Engagement of threaded parts. All threaded external parts shall engage by at least three full threads.

3.4.3 Case. The case shall not be electrically connected to the contacts or coil; however, it may be used as part of the magnetic circuit.

3.4.4 Contacts. Contacts shall have the load ratings and arrangements as specified (see 3.1), and shall be capable of carrying the maximum rated current continuously.

3.4.5 Coils. Coils shall be adequately insulated electrically from the frame, the contacts, and the enclosure and shall be designed for continuous operation at rated voltage. The resistance and rated voltage shall be as specified (see 3.1).

3.4.6 Terminal identification. Terminals shall be identified as specified (see 3.1).

3.4.7 Circuit diagram. The circuit diagram shall be as specified (see 3.1).

3.4.8 Temperature range. The relay operational temperature range shall be as specified (see 3.1).

3.5 Solderability (applicable to solder terminals when specified). When relays are tested as specified in 4.7.2, the dipped surface of solid wire-lead and pin terminals shall be at least 95 percent covered with a continuous new solder coating. The remaining 5 percent may contain only small pinholes or rough spots; these shall not be concentrated in one area. Bare base metal where the solder dip failed to cover the original coating is an indication of poor solderability, and shall be cause for failure. For solder-lug terminals greater than 0.045 inch (1.14 mm) in diameter, 95 percent of the total length of fillet, which is between the standard wrap wire and the terminal, shall be tangent to the surface of the terminal being tested, and shall be free of pinholes, voids, etc. A ragged or interrupted line at the point of tangency between the fillet and the terminal under test shall be considered a failure.

3.6 Insulation resistance. When relays are tested as specified in 4.7.3, the insulation resistance shall be 10,000 megohms or more, unless otherwise specified (see 3.1).

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3.7 Dielectric withstanding voltage. When relays are tested as specified in 4.7.4 they shall withstand the specified voltage without damage. The leakage current shall not exceed 1 mA and there shall be no evidence of damage due to arcing (air discharge), flashover (surface discharge), or insulation breakdown (puncture discharge). After high level tests, the dielectric withstanding voltage should be at least 75 percent of the initial sea level value (see 3.1).

3.8 Electrical characteristics.

3.8.1 Static contact resistance (or voltage drop). When relays are tested as specified in 4.7.5.1, the static contact resistance (or voltage drop) of any pair of mated contacts shall not exceed the applicable values specified in table I.

TABLE I. Static contact resistance (or voltage drop).

Contact rating or relay type	Load conditions	Maximum allowable static contact resistance (or voltage drop)	
		Before life	During and after life
High level (more 2 amp but not in excess of 10 amp)	Rated current at 6 or 28 V dc or peak ac	100 mV	200 mV
High level (2 amp or less)	Rated current at 6 or 28 V dc or peak ac	0.05 ohm	0.10 ohm
Intermediate current	100 ±5 mA at 28 V dc or peak ac	0.05 ohm	Level 1-1.0 ohm; Level 2-3.0 ohms
Low level	10 mA maximum at 6 V open-circuit (maximum dc or peak ac)	0.05 ohm	0.15 ohm

3.8.2 Pickup, hold, and dropout voltage (or current). When relays are tested as specified in 4.7.5.2, the pickup, hold, and dropout voltage 1/ shall be as specified (see 3.1).

3.8.3 DC coil resistance. When relays are tested as specified in 4.7.5.3, the dc coil resistance shall be as specified (see 3.1).

3.8.4 Operate and release time (including bounce). When relays are tested as specified in 4.7.5.4 the operate and release time shall be as specified (see 3.1).

3.9 Shock (specified pulse) (nonoperating) (when specified see 3.1). When relays are tested as specified in 4.7.6, there shall be no evidence of mechanical or electrical damage.

3.10 Vibration (nonoperating) (when specified see 3.1). When relays are tested as specified in 4.7.7, there shall be no evidence of breaking, cracking, chipping or flaking of the finish, or loosening of the terminals.

1/ Wherever "voltage" is used herein, it is considered to include "current" if applicable.

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3.11 Moisture resistance (when specified) (see 3.1). When relays are tested as specified in 4.7.8, there shall be no evidence of breaking, cracking, chipping or flaking of the finish, or loosening of the terminals. Immediately after step 6 of the final cycle, the insulation resistance shall be 100 megohms or more. After the 24-hour drying period, the dielectric withstanding voltage, insulation resistance, static contact resistance, and pickup and dropout voltages shall be as specified in 3.6, 3.9, 3.13, and 3.11, respectively.

3.12 Resistance to soldering heat (applicable to solder terminal when specified). When relays are tested as specified in 4.7.9 there shall be no damage which would adversely affect the operation of the relay.

3.13 Terminal strength. When relays are tested as specified in 4.7.10, there shall be no evidence of loosening or breaking of the terminals; and there shall be no deformation to the threads of screw terminals; no bending (see NOTE) of a pin or damage to the insulating base of plug-in relays; nor shall there be any other damage which would adversely affect the normal operation of the relay.

NOTE: Bending of solder terminals shall not be construed as damage; bending of plug-in terminals shall not be construed as damage provided they can be re-formed in a manner to permit proper mating with the applicable socket.

3.14 Magnetic interference (when specified, see 3.1). When relays are tested as specified in 4.7.11, the pickup and dropout values shall be within the limits specified (see 3.1).

3.15 Sand and dust. When relays are tested as specified in 4.7.12, there shall be no evidence of damage sufficient to impair the operation of the relay.

3.16 Contact sticking (when specified, see 3.1). When relays are tested as specified in 4.7.13 the release time shall be as specified (see 3.1).

3.17 Humidity (steady state). When relays are tested as specified in 4.7.14, there shall be no degradation of electrical and dielectric characteristics. No swelling of materials that destroy functional utility, no cracking, peeling, flaking, chipping, crazing, or other impairment of the protective coating shall be in evidence.

3.18 Intermediate current (applicable to high level relays only). When relays are tested as specified in 4.7.15, during cycling, unless otherwise specified, the dynamic contact resistance of a closed contact shall not exceed the applicable value specified in table V and the voltage across an open contact shall be 95 percent or more applied load voltage. After cycling, the closed static contact resistance shall not exceed the limits specified in table VI.

3.19 Overload (applicable to high level relays only). When relays are tested as specified in 4.7.16, the voltage drop across closed contacts shall be not more than 5 percent of applied circuit load voltage and the voltage across open contacts shall be not less than 95 percent of applied circuit voltage. The case-to-ground fuse shall remain electrically continuous.

3.20 Life. When relays are tested as specified in 4.7.17 during cycling, the contact miss detector's monitoring level shall be less than 100 ohms for low level testing and less than 5 percent of the open-circuit voltage for high level testing and the voltage across open contacts shall be 95 percent or more of the open circuit voltage. The case to ground fuse shall remain electrically continuous. The static contact resistance during and following cycling shall not be greater than twice the initial specified contact resistance (or voltage drop) except low level (see table VI).

3.21 Marking (see 3.1). Relays shall be marked in accordance with MIL-STD-1285, with the following information:

a. Military part number (see 1.2).

b. Rated coil voltage (see 3.4.5 and 3.1) and when applicable, operating frequency.

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c. Coil resistance (see 3.4.5 and 3.8.3).

d. Contact rating (the highest dc resistive load rating shall be marked when plural loads are specified (see 3.4.4 and 3.1).

e. Circuit diagram (see 3.4.7 and 3.1).

f. Terminals (see 3.4.6 and 3.1).

g. Source and date codes in accordance with MIL-STD-1285.

3.22 Workmanship. The relays shall be fabricated in such a manner as to be uniform in quality, and shall be free from cracked or displaced parts, sharp edges burrs, and other defects that will affect life, serviceability, and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The contractor shall establish and maintain a calibration system in accordance with MIL-STD-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.3).

b. Qualification inspection (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table II, used in fabricating the relays, are in accordance with the applicable referenced specifications prior to such fabrication.

TABLE II. Materials inspection.

Material	Requirement paragraph	Applicable specification
Ceramic - - - - - - - -	3.3.2	MIL-I-10
Magnet wire - - - - - -	3.3.3	J-W-1177
Rubber- - - - - - - - -	3.3.6	ZZ-R-765

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the GENERAL REQUIREMENTS of MIL-STD-202.

4.4.1 Power supply. Unless otherwise specified herein, the power supply shall have no more than 10 percent regulation at 110 percent of the specified test load current. A dc power supply shall have no more than 5 percent voltage ripple. An ac power supply shall be within 1 percent of the specified frequency and shall be sinusoidal with a form factor between 0.95 and 1.25.

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4.4.2 Grounding. Unless otherwise specified (see 3.1), the negative side of the dc power supply shall be grounded; one side of single-phase ac power supply shall be grounded; or the neutral of 3-phase ac power supply shall be grounded, as applicable.

4.4.3 Load conditions during tests. The coil(s) of the relay being tested shall be connected to the grounded side of their power supply; the loads of the relay being tested shall be connected to the grounded side of their power supply; and each stationary contact shall be connected to an individual load. All tests during which the contacts are loaded and being cycled, except dielectric withstanding voltage, shall be conducted with the case of the relay connected to the power supply grounded or neutral through a normal blow fuse rated at 5 percent of the contact load, maximum, but not less than 0.100 ampere. For relays with nongrounded case ratings, tests for isolated-case ratings may be made with the case electrically isolated from the power supply ground.

4.4.4 Testing devices. Devices used in the testing of low level relays shall not load the contacts above 10 mA resistive at 6 V dc or peak AC maximum open circuit unless otherwise specified herein. High level contacts shall not be loaded above their lowest rating.

4.4.5 Mounting relays for ambient temperature tests. When relays are subjected to the tests specified in 4.7.15, (intermediate current), they may be mounted on a heat sink in accordance with the following:

a. Each relay may be attached by its normal mounting means to a 1/16 inch thick, minimum flat aluminum plate heat sink. The heat sink shall be designed to place every relay in the center of its own square space whose total surface area (both sides) is eight times the outside surface area of ther relay, excluding mounting. Relays without mounts shall be held to the heat sink with a metal strap 1/4 inch wide by 0.015 inch maximum thickness. The heat sink assembly shall be suspended by twine or other nonheat conducting material in a plane parallel to the normal air flow in the oven. The leads shall not constitute a heat sink.

b. Chamber temperature shall be controlled to maintain the temperature at the specified ambient extremes (see 3.1).

4.4.6 Methods of examination and test. Application of coil power to relays under test shall be such that plus polarity is applied to the color coded terminal when applicable; or to the lower numbered terminal when color coding is not used. Testing of dual-coil relays shall be repeated with each coil serving as the operating coil; and testing of latching relays shall be repeated with the relay in each operated position. When applicable, testing of relays with plug-in termination shall be performed with the appropriate or specified socket or connector mated to the relay.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.5.1 Sample size. The number of relays to be subjected to qualification inspection shall be as specified in table III.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in the table III in the order shown. With the exception of the one unsealed sample unit which is to be examined internally, all sample units shall be subjected to the inspections of group I. The sample shall then be divided as specified in table III for groups II through IV (as applicable), and subjected to the inspections for their particular groups.

4.5.3 Failures. Failures in excess of those allowed in table III shall be cause for refusal to grant qualification approval.

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TABLE III. Qualification inspection.

Inspection	Requirement	Test method paragraph	Number of sample units for inspection	Number of defectives permitted
Group I				
Visual and mechanical inspection:				
Internal 1/	3.1,3.3,3.4, and 3.22	4.7.1		
External	3.1,3.3,3.4, 3.21 and 3.22	4.7		
Solderability (when specified) (applicable to solder terminals) (3 sample units)	3.5	4.7.2		
Insulation resistance	3.6	4.7.3	All sample units	1
Dielectric withstanding voltage	3.7	4.7.4		
Electrical characteristics:				
Static contact resistance (or voltage drop)	3.8.1	4.7.5.1		
Pickup, hold, and dropout voltage)	3.8.2	4.7.5.2		
DC coil resistance	3.8.3	4.7.5.3		
Operate and release time (including bounce)	3.8.4	4.7.5.4		
Group II				
Shock (when specified)	3.9	4.7.6		
Vibration (when specified)	3.10	4.7.7		
Moisture resistance (when specified)	3.11	4.7.8		
Insulation resistance	3.6	4.7.3		
Dielectric withstanding voltage	3.7	4.7.4		
Electrical characteristics:				
Static contact resistance (or voltage drop)	3.8.1	4.7.5.1	4	0
Pickup, hold, and dropout voltage	3.8.2	4.7.5.2		
DC coil resistance	3.8.3	4.7.5.3		
Operate and release time (including bounce)	3.8.4	4.7.5.4		
Visual and mechanical inspection	3.1,3.3,3.4, 3.21 and 3.22	4.7.1		
Group III				
Resistance to soldering heat (applicable to solder terminals) (when specified)	3.12	4.7.9		
Terminal strength 2/	3.13	4.7.10		
Magnetic interference (when specified)	3.14	4.7.11		
Sand and dust	3.15	4.7.12		
Insulation resistance	3.6	4.7.3		
Dielectric withstanding voltage	3.7	4.7.4		
Electrical characteristics:				
Static contact resistance (or voltage drop)	3.8.1	4.7.5.1	4	0
Pickup, hold, and dropout voltage	3.8.2	4.7.5.2		
DC coil resistance	3.8.3	4.7.5.3		
Operate and release time (including bounce)	3.8.4	4.7.5.4		
Visual and mechanical inspection	3.1,3.3,3.4, 3.21 and 3.22	4.7.1		

See footnotes at end of table.

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TABLE III. Qualification inspection - Continued.

Inspection	Requirement	Test method paragraph	Number of sample units for inspection	Number of defectives permitted
Group IV				
Contact sticking (when specified)	3.16	4.7.13	4 relays, minimum. For relays with more than 2 contact ratings, 2 relays per contact rating shall be tested.	0
Humidity (steady state)	3.17	4.7.14		
Intermediate current (applicable to high level relays only)	3.18	4.7.15		
Overload (applicable to high level relays only)	3.19	4.7.16		
Life	3.20	4.7.17		
Insulation resistance	3.6	4.7.3		
Dielectric withstanding voltage	3.7	4.7.4		
Electrical characteristics:				
Static contact resistance (or voltage drop)	3.8.1	4.7.5.1		
Pickup, hold, and dropout voltage	3.8.2	4.7.5.2		
DC coil resistance	3.8.3	4.7.5.3		
Operate and release time (including bounce)	3.8.4	4.7.5.4		
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.21 and 3.22	4.7.1		

1/ One sample unit shall be subjected to the internal inspection.

2/ Test four sample relays which have been subjected to the highest current loads.

4.5.4 Retention of qualification. To retain qualification, the contractor shall forward a report to the qualifying activity at 12-month intervals. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery (group A), indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

b. The results of tests performed for periodic inspection, group B, including the number and mode of failures. The test report shall include results of all periodic inspection tests performed and completed during the 36-month period. If the test results indicate nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 60 days after the end of each 36-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 36-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification (see 4.6.2.1).

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of each part number to testing in accordance with the qualification inspection requirements.

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4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection. Except as specified in 4.7.2.1.3, delivery of products which have passed group A inspection shall not be delayed pending the results of the group B inspections.

4.6.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of all relays of the same enclosure, configuration, vibration characteristic, temperature class, shock type, and terminal type, as applicable and when specified, produced under essentially the same conditions within a period not to exceed 4 weeks, and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table IV.

4.6.1.2.1 Sampling plan (group A inspection). Subgroup 1 and 2 tests shall be performed on each relay offered for inspection.

4.6.1.2.2 Rejected lots (group A inspection).

a. Subgroup 1. If over 10 percent of an inspection lot fails, the relays contained in that lot shall be rejected and shall not be resubmitted for inspection. Relays which have failed to pass subgroup 1 tests to which they have been subjected, shall be removed from the lot and discarded.

b. Subgroup 2. Rejected relays may be reworked to correct defects and resubmitted for inspection.

4.6.1.2.3 Disposition of sample units. Sample relays which have passed all the group A inspections may be delivered on the contract or purchase order, if the lot is accepted and the sample relays are still within the specified electrical tolerances.

4.6.2 Periodic inspection. Periodic inspection shall consist of group B.

4.6.2.1 Group B inspection. Group B inspection shall consist of the inspections specified in table III, and shall be performed in the order shown. A summary of group B tests indicating the number of lots accepted and the number rejected shall be submitted to the qualifying activity at 36-month intervals.

TABLE IV. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	
Subgroup 1 1/			
Insulation resistance	3.7	4.7.3	100% inspection discard all failed relays
Dielectric withstanding voltage	3.7	4.7.4	
Electrical characteristics:			
Static contact resistance (or voltage drop)	3.8.1	4.7.5.1	
Pickup, hold, and dropout voltage	3.8.2	4.7.5.2	
DC coil resistance	3.8.3	4.7.5.3	
Operate and release time (including bounce)	3.8.4	4.7.5.4	
Subgroup 2			
Visual and mechanical inspection	3.1,3.3,3.4, 3.21 and 3.22	4.7.1	
External 2/			

1/ The order of performing the inspections is optional.
2/ Physical dimensions and weight shall be checked on only two relays selected at random from each lot.

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4.6.2.1.1 Sampling plan. Every 36 months, the number of sample units specified in 4.5.1, shall be subjected to the inspections of table III, and shall be performed in the order shown. The number of failures allowed shall be as specified in table III.

4.6.2.1.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.6.2.1.3 Noncompliance. If a sample fails to pass the group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Group A inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6.3 Inspection of packaging. The sampling and inspection of the preservation and interior pack marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Methods of inspection.

4.7.1 Visual and mechanical. Relays shall be examined to verify that the materials, external design and construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (sees 3.1, 3.3, 3.4, 3.21 and 3.22).

4.7.2 Solderability (applicable to solder terminals) (when specified see 3.5). Relays shall be tested in accordance with method 208 of MIL-STD-202. The following detail and exception shall apply:

a. Number of terminations of each part to be tested - All.

b. The temperature of the molten solder shall be a uniform 260° ±5°C (500° ±9°F).

4.7.3 Insulation resistance (see 3.6). Relays shall be tested in accordance with method 302 of MIL-STD-202 with the relay in the energized and de-energized positions. The following details shall apply:

a. Test-condition letter - A or B (for relays with coil and contact ratings below 60 volts), and B (for all other relays).

b. Points of measurement - As specified in points of application in table V.

4.7.4 Dielectric withstanding voltage (see 3.7). Relays shall be tested as specified in 4.7.4.1, and when specified (see 3.1) in accordance with 4.7.4.2. Testing in accordance 4.7.4.2 is not required for group A testing.

4.7.4.1 At atmospheric pressure. Relays shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Points of application and magnitude of test voltage - As shown in table V.

b. Maximum leakage current - 1.0 mA.

c. Duration of application - 60 seconds for qualification and group B tests; 5 seconds for group A tests.

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TABLE V. Test details for dielectric withstanding voltage.

Points of application	Test voltage
Between case, frame, or enclosure, and between open contacts in the energized and de-energized position - -	1,000 volts ac plus twice rated voltage or as specified (see 3.1)
Between case, frame, or enclosure and coil(s)- - - - - -	
Between all contacts and coil(s) - - - - - - - - - - - -	
Between open contacts in the energized and de-energized position- -	
Between coils of dual-coil relays- - - - - - - - - - - -	
Between contact poles- - - - - - - - - - - - - - - - - -	

4.7.4.2 At reduced barometric pressure. Relays specified for operation above 10,000 feet shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

a. Method of mounting - Normal mounting means.

b. Test-condition letter - C (unless otherwise specified, see 3.1).

c. Tests during subjection to reduced pressure - As specified in 4.7.4.1, except test voltage shall be 350 volts.

Following these tests, relays shall be examined for evidence of arcing, flashover, insulation breakdown, and damage.

4.7.5 Electrical characteristics.

4.7.5.1 Static contact resistance (or voltage drop) (see 3.8.1). Relays shall be tested in accordance with method 307 of MIL-STD-202. The following details and exception shall apply:

a. Method of connection - Between the measuring appartus and the relay terminals (for relays with wire-lead terminals, this measurement shall be made 1/2 ± 3/32 inch from the emergence of the lead from the relay). Voltage-sensing leads shall be connected in such a way so as to exclude the resistance of the current-carrying leads.

b. Test current - As specified in table VI.

c. Maximum open-circuit test voltage - As specified in table VI.

d. Points of measurements - All contacts in their closed position; the coil shall be energized with rated voltage (or current) if necessary to effect contact closure.

e. Number of actuations prior to measurement - None.

f. Number of test actuations - Three (no contact voltage shall be applied during contact transfer). For quality conformance inspection, one test actuation shall be made.

g. Number of measurements per actuation - One in each closed contact position.

4.7.5.2 Pickup, hold, and dropout voltage (see 3.8.2). Pickup, hold, and dropout voltage shall be measured as specified in 4.7.5.2.1, 4.7.5.2.2, 4.7.5.2.3, or 4.7.5.2.4. For qualification inspection, the relay shall be mounted in each of three mutually perpendicular planes, unless otherwise specified (see 3.1). For quality conformance inspection, the relay mounting position is optional. A suitable indicating device shall be used to monitor contact position. During the test, all contacts shall not change state (break or remake) when the coil is energized above the pull-in voltage, when the coil voltage is lowered to any value above the specified hold (maximum dropout) voltage, or when the coil voltage is reduced from (minimum) dropout voltage to zero.

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4.7.5.2.1 Pickup voltage (not applicable to latching relays). Rated voltage shall be applied to the coil for a period of 1 to 3 seconds, then reduced to zero. The voltage shall then be gradually increased until the relay operates, and the pickup voltage shall be measured. For group A inspection, it is permissible to perform pickup and dropout voltage using step function voltage changes as follows and as illustrated on figure 1.

a. Apply rated coil voltage.

b. Stepdown to hold voltage, normally open contacts must still be making.

c. Stepdown to dropout voltage, all contacts should have transferred and all normally closed contacts should be making.

d. Stepdown to zero voltage.

e. Stepup to the maximum pickup voltage, contacts should have transferred and all normally open contacts should be made.

4.7.5.2.2 Pickup (operate) voltage (applicable to latching relays only). Before measuring the pickup voltage, establish that all contacts are in their last energized mode for dual coil relays or in the plus (+) contrasting color bead mode for single coil relays (see 3.1). If not, apply rated voltage to establish last energized contact position. Gradually increase the voltage to the latching coil or in the latching direction for single coil relays until the contacts transfer and the latch (operate) voltage measured. Apply rated latching voltage and reduce to zero. Gradually increase the voltage to the reset coil or in the reset direction for single coil relay until the contacts transfer and the reset (operate) voltage measured. For group A inspection, it is permissible to perform this operate voltage measurement using step function voltage changes illustrated on figure 1.

4.7.5.2.3 Hold voltage. Rated voltage shall be applied to the coil. The voltage shall then be reduced to the specified hold voltage (see 3.1). All contacts shall not change state. For group A inspection, it is permissible to perform this test using the step function voltage program described in 4.7.7.2.1 and on figure 1.

4.7.5.2.4 Dropout voltage (not applicable to latching relays). Rated voltage shall be applied to the coil. The voltage shall then be gradually reduced until all contacts return to the de-energized position and the dropout voltage shall be measured. For group A inspection, it is permissible to perform this test using the step function voltage program described in 4.7.7.2.1 and on figure 1.

4.7.5.3 DC coil resistance (see 3.8.3). Relays shall be tested in accordance with method 303 of MIL-STD-202.

4.7.5.4 Operate and release time, including bounce (see 3.8.4). Operate and release time shall be measured using an oscilloscope or other acceptable means approved by the qualifying activity. Rated voltage shall be applied to the coil. Contact load conditions shall be as required in 4.7.8.6. The circuit shown on figure 2, or equivalent, shall be used. The operate and release time shall be exclusive of contact bounce. Timing measurements shall be made on all contact sets, electronic instrumentation or other suitable means may be used for group A inspection.

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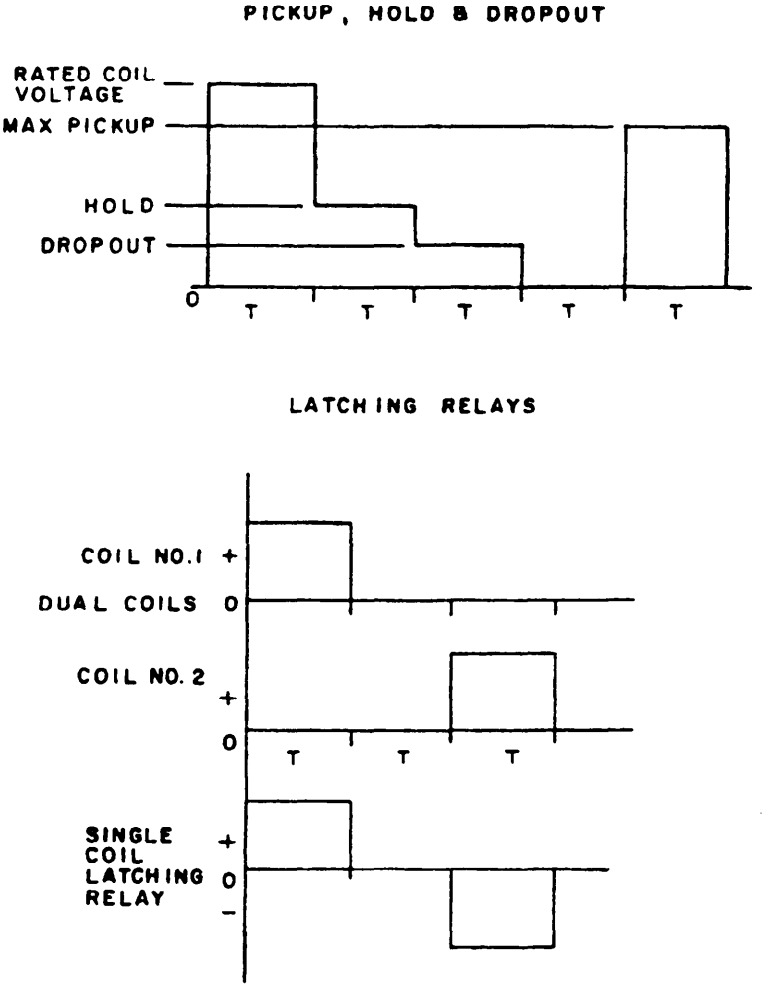
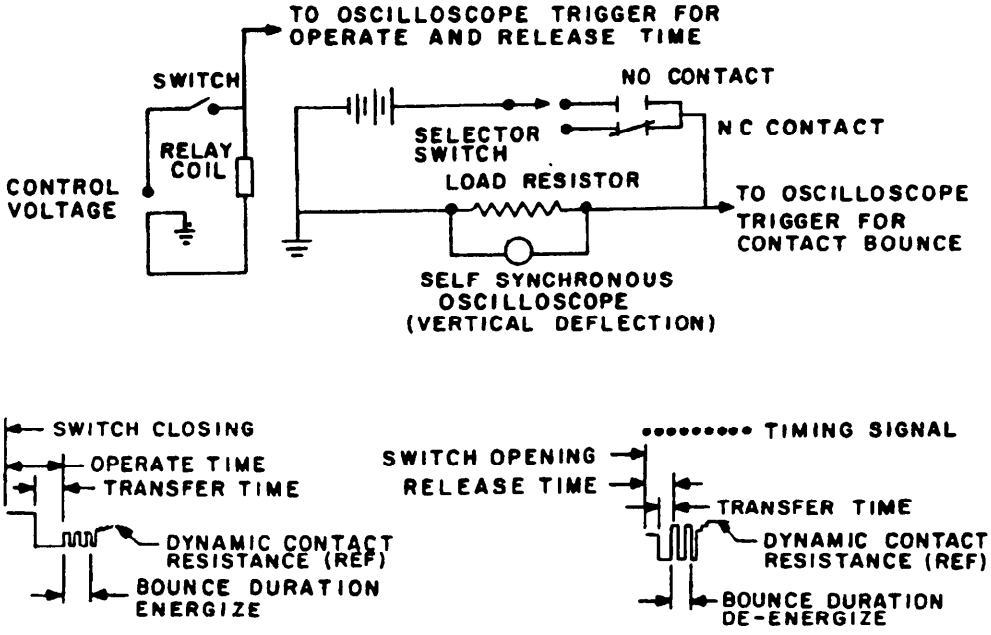


FIGURE 1. Optional pickup, hold, and dropout sequencing for group A inspection.

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NOTE: The horizontal scan rate of the oscilloscope shall show the required pertinent data.

FIGURE 2. Typical circuit for operate and release time (including bounce) with typical traces.

4.7.6 Shock (specified pulse, nonoperating) (when specified) (see 3.9). Relays shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting method - Rigidly mounted by normal mounting means. (Relays designed without mounting provisions shall be rigidly secured to a suitable nonmagnetic mounting plate by means of potting or adhesive applied between the top of the relay and the mounting plate.

b. Test condition - A, except peak value shall be 50 G's.

c. Electrical load conditions - The coil shall be deenergized and the contacts shall not be loaded or monitored for chatter and closure.

d. Examination often test - Relays shall be examined for evidence of structural failure or other damage which might impair the operation of the relay.

4.7.7 Vibration (nonoperating) (when specified) (see 3.10). Relays shall be tested in accordance with method 204 of MIL-STD-202. The contacts shall not be loaded or monitored for chatter or closure during vibration. The following details shall apply:

a. Mounting - Same as 4.7.6(a).

b. Test condition - D, except amplitude shall not exceed 20 G's and frequency range shall be 10 to 2,000 Hz.

c. Examinations - Same as 4.7.6(d).

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4.7.8 Moisture resistance (when specified, see 3.1 and 3.11). Relays shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting - On a corrosion-resistant panel by normal mounting means.

b. Initial measurement - Not applicable.

c. Loading voltage - Not applicable.

d. Final measurements - Upon completion of step 6 of the final cycle, insulation resistance shall be measured as specified in 4.5.6. After a 24-hour drying period at a relative humidity of 50 ±5 percent, dielectric withstanding voltage shall be measured as specified in 4.7.4, except the test voltage shall be 90 percent of the initial potential. The insulation resistance, static contact resistance, and pickup, hold, and dropout voltage shall be measured as specified in 4.7.3, 4.7.5.1, and 4.7.5.2 respectively.

e. Examination after test - Relays shall be examined for evidence of breaking, cracking, chipping or flaking of the finish, or loosening of the terminals.

4.7.9 Resistance to soldering heat applicable to solder terminals (when specified, see 3.1) (see 3.12). Relays shall be tested in accordance with method 210 of MIL-STD-202. The following details and exception shall apply:

a. Depth of immersion in molten solder - Within .060 ±.020 inch of the relay base.

b. Test-condition letter - B.

c. Examination after test - As specified in 4.7.6(d).

4.7.10 Terminal strength (see 3.13). Relays shall be tested in accordance with method 211 of MIL-STD-202 and in accordance with the following, as applicable. Unless otherwise specified herein, two terminals of each discrete design, size, and configuration shall be tested; however, if there is only one of such design, size, and configuration, it shall be tested.

4.7.10.1 Pull test (all terminal types). Terminals shall be tested as specified in test-condition letter A, the force shall be as specified (see 3.1).

4.7.10.2 Bend test (not applicable to plug-in terminals). Terminals shall be tested as specified in test-condition letter B (two bends) or C, as applicable. Loads for test-condition letter C shall be as specified (see 3.1).

4.7.10.3 Bend test (plug-in terminals of a standard octal base only). The lesser of five or all terminals shall be tested by applying a force of 5 pounds perpendicular to the axis of the pin within 1/8 inch of the tip of the pin.

4.7.10.4 Twist test (wire lead terminals only). All terminals shall be tested as specified in test-condition letter D, except during application of torsion, each terminal shall be rotated 45 degrees in one direction, then returned to start; rotated in opposite direction 45 degrees, then returned to start. Each terminal shall be subjected to two such rotations and returns. Each terminal shall be held at a point 3/4 inch from the point of emergence from the relay and in one plane shall be bent 20 ±5 degrees in one direction, then returned to start; rotated in opposite direction 20 ±5 degrees, then returned to start; this procedure shall then be repeated in the perpendicular plane.

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4.7.10.5 Torque test (screw terminals only). All terminals shall be tested as specified in test-condition letter E, except as follows:

Screw size	Torque pound-inches
4-40	4.4
6-32	10.0
8-32	20.0
10-32	32.0
10-24	35.0

Following these tests, relays shall be examined for evidence of loosening or breaking of the terminals and other damage that could adversely affect the normal operation of the relay.

4.7.11 Magnetic interference (when specified, see 3.1) (see 3.14). Relays shall be tested in accordance with 4.7.15.1 or 4.7.15.2, as specified (see 3.1).

4.7.11.1 Uniform-stray-field. The relay shall be mounted by suitable nonmagnetic means within the central volume of the test coil (see 6.6). The critical axis of the relay shall be alined with the longitudinal axis of the test coil. Pickup, hold, and dropout voltage (or current) shall be measured as specified in 4.7.5.2 in 0 magnetic field and in 100 gauss magnetic field of both polarities.

4.7.11.2 Adjacent-similar-relay. The relay under test and eight similar relays shall be mounted in the same physical orientation by nonmagnetic means, as shown on figure 4. Unless otherwise specified, grid-spaced relays shall be mounted so that all terminals are positioned in the closest possible grid pattern. The spacing between terminals and tops of adjacent terminals shall be as shown on figure 3, unless otherwise specified. Pickup, hold, and dropout voltage (or current) shall be measured on the relay under test as specified in 4.7.5.2 with the coils of the eight outer relays energized at rated voltage (the magnetic polarity of each relay shall be similarly oriented) (see 6.6). This measurement shall be repeated with the coils of the eight outer relays de-energized.

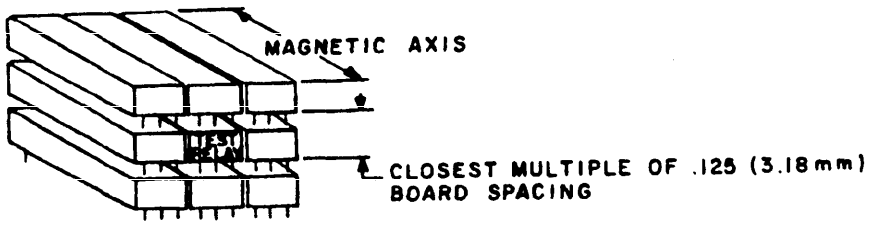


FIGURE 3. Mounting array for adjacent similar relays.

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4.7.12 Sand and dust (see 3.15). Relays shall be tested in accordance with method 110 of MIL-STD-202. The following details and exception shall apply:

a. Test-condition letter - A (except during part II, the temperature shall be the maximum temperature rating of the relay, see table III).

b. Measurements after test - Immediately after reaching room temperature, static contact resistance (or voltage drop) and pickup, hold, and dropout voltage (or current) shall be measured as specified in 4.7.5.1 and 4.7.5.2, respectively.

4.7.13 Contact sticking (when specified, see 3.1) (see 3.16). Relays shall be energized for 96 hours with 120 percent of rated voltage applied to the coil. No load shall be applied to the contacts. At the end of this period, without physically disturbing the relay, the coil shall be de-energized and the release time shall be measured as specified in 4.7.5.4.

4.7.14 Humidity (steady state) (see 3.17). Relays shall be tested in accordance with method 103 of MIL-STD-202. The following details shall apply:

a. Measurements after conditioning - Not applicable.

b. Test condition letter - B, unless otherwise specified (see 3.1).

c. Loading voltage - Not applicable.

d. Examinations - See 4.7.6(d).

4.7.15 Intermediate current (applicable to high level relays only) (see 3.18). For qualification inspection, relays shall be tested in accordance with 4.7.15.1 specified (see 3.1). Relays shall be mounted as specified in 4.4.5. The ambient temperature of the test chamber shall be the maximum temperature rating of the relay. Unless otherwise specified (see 3.1), coils shall be energized with rated voltage at the rate specified herein; however, intermittent duty and pulse-operated relays shall be operated as specified (see 3.1). The same type load shall be used for each contact of the same pole. Contacts shall be monitored for dynamic contact resistance at each operation; however, unless other-wise specified (see 3.1), contacts carrying rated resistive load need not be monitored. Unless otherwise specified (see 3.1), contacts shall be loaded as follows; however, if any load specified herein exceeds the contact rating (see 3.1), the load shall be as specified (see 3.1).

First pole - 0.1 ampere resistive at 28 ±1 volts dc.

Second pole - 0.3 ampere inductive (220 millihenries) at 28 ±1 volts dc.

Third pole - 0.5 ampere resistive at 28 ±1 volt dc.

Fourth pole - Rated resistive at 28 ±1 volts dc.

NOTE: Single-pole relays shall not be tested at rated resistive load and the other three loads shall be distributed throughout the test sample in the order shown. Two-pole and three-pole relays shall have one pole loaded with rated resistive load and the other three loads shall be distributed throughout the test sample in the order shown. Relays having more than four poles shall alternate the 0.1 ampere resistive and rated loads on the remaining poles. For group B inspection, relays shall be tested as specified herein, except the operations shall be 10,000.

4.7.15.1 Number of operations. Relays shall be subjected to 100,000 operations (unless otherwise specified, see 3.1), at the rate of 20 ±2 cycles per minute, with "on" and "off" times approximately equal.

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4.7.16 Overload (applicable to high level relays only) (see 3.19). Relay contacts (both normally open and normally closed) shall be subjected to the loads specified in 4.7.8.16.3, as applicable. All normally open contacts shall be tested together, then all normally closed contacts tested together. The number of operations shall be 100 for dc contact loads, and 200 for ac contact loads ("on" and "off" times shall be approximately equal). The coil shall be energized at rated voltage. Overload current shall be twice rated load current. The opening and closing of all contacts shall be continuously monitored to detect the actual mechanical and electrical switching of the contacts. The equipment shall be capable of monitoring 50 percent of the actual opened and closed time of each contact. The equipment shall automatically cut off when a failure occurs, or shall record any failures. The relay frame, case, or enclosure shall be connected to system-ground through a normal-blow fuse rated at 5 percent of the test load or 100 mA, whichever is greater. After the test, the fuse shall be tested for electrical continuity.

4.7.16.1 Resistive. Suitable resistors shall be used. Cycling rate shall be 20 ±2 cycles per minute (c/m).

4.7.16.2 Inductive. Cycling rate shall be 10 ±1 c/m.

4.7.16.2.1 Inductive, dc. Unless otherwise specified (see 3.1), inductive dc loads shall be computed in accordance with the formula $CE^2 = (0.28)\ I^{1.18}$. A shunting capacitor shall be placed across the test contacts to absorb the arc energy. The voltage across the capacitors shall be measured upon circuit interruption by means of an oscilloscope and shall be taken as the peak value of the first oscillation.

Where: W = energy (in joules).
C = capacitance of shunt capacitor (in farads).
E = voltage across capacitor (in volts).
I = maximum dc inductive current rating of the contacts.

The energy thus calculated is the energy which would be dissipated by the contacts if the capacitor were removed. This energy shall be within 10 percent of the energy calculated by the formula $W = (0.14)I^{1.18}$. This method of energy measurement requires the use of a pulse-type noninductive capacitor having a working voltage of 1,000 volts. The size of the capacitor shall be such that the peak voltage measured shall be not less than 200, nor greater than 900 volts. During the overload test, the capacitor shall be disconnected. When specified (see 3.1), the inductive load shall have an L/R ratio of 0.008.

4.7.16.2.2 Inductive, ac. The load shall consist of inductive and resistive elements with 0.7 ±0.05 lagging power factor at the voltage and frequency specified (see 3.1).

4.7.16.3 Lamp load. Unless otherwise specified, the lamp load shall be performed with the 28 Vdc power supply voltage. Relays shall make 12 times the rated lamp load and break the rated lamp load. The duration of the 12 times inrush shall be 0.015-0.020 second, the total "on" time shall be 2 ±0.05 seconds and the "off" time shall be 7 ±2 seconds.

4.7.17 Life (see 3.20). Unless otherwise specified (see 3.1), relays shall be cycled for 100,000 operations with contacts loaded in accordance with 4.7.17.1 through 4.7.17.5 as applicable. The relay coil energization shall be nonsynchronous with the power supply for ac loads. The relay case, cover, or enclosure shall be connected to the system-ground by a normal-blow fuse rated at 5 percent of the test current or 100 mA, whichever is greater. This test shall be performed at the maximum ambient temperature of the relay (see table III). With the exception of motor loads, opening and closing of all contacts shall be continuously monitored for dynamic contact resistance under low level load conditions (see table VI). The equipment shall be capable of testing 50 percent of the actual closed time of the contact. This equipment shall automatically cut off when a failure occurs, or shall record any failures. After the test, the fuse shall be tested for electrical continuity.

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4.7.17.1 Resistive load. Cycling rate shall be 20 ±2 operations per minute. Suitable resistors shall be used. Current shall be rated resistive current as specified (see 3.1). When specified (see 3.1), following the 100,000 operations at rated resistive current, the relay shall be cycled for an additional 400,000 operations with the current reduced to 25 percent of the rated resistive current.

4.7.17.2 Inductive load. Current shall be rated current. Appropriate inductive load components (see 4.7.16.2.1) shall be used. A suitable resistor may be placed in the circuit to obtain rated steady-state current flow. Cycling rate shall be 10 ±1 c/m with equal "on" and "off" periods.

4.7.17.3 Lamp load. The load and cycling rate shall be as specified in 4.7.16.3.

4.7.17.4 Motor load. Unless otherwise specified (see 3.1), resistors shall be used. Resistive load current shall be five times rated motor load current (see 3.1). Cycling rate shall be 2 to 3 c/m with equal "on" and "off" times.

4.7.17.5 Low level. Relays shall be tested in accordance with method 311 of MIL-STD-202. The following details shall apply:

a. Number of misses which will be considered a failure - One.

b. Maximum contact resistance allowed - As specified in tables II and VI, as applicable.

c. Number of cycles of operation and cycling rate - 100,000 cycles, unless otherwise specified (see 3.1) at the rate of 60 to 300 cycles per minute (c/m).

d. Contact load - As specified in table VI.

TABLE VI. Dynamic contact resistance (or voltage drop).

Contact rating or relay type	Load conditions	Maximum allowable dynamic contact resistance (or voltage drop)
High level	Rated current at rated voltage	10% open-circuit voltage
Intermediate current	See 4.7.15	3 ohms
Low level	10 to 50 µA at 10 to 50 mV open-circuit (maximum dc or peak ac)	100 ohms

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Relays shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Relays shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation application. Preservations shall not be used.

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5.1.1.4 Unit packs. Relays shall be individually unit packed in accordance with MIL-P-116, submethod IA-8, insuring compliance with the applicable requirements of that specification. Each unit pack shall be placed in a supplementary container conforming to PPP-B-566 or PPP-B-676.

5.1.1.5 Intermediate packs. Intermediate packs are not required.

5.1.2 Level C. The level C preservation of relays shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The packaged relays shall be packed in wood boxes conforming to PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-585, class 3. Closure and strapping shall be in accordance with the applicable container specification.

5.2.2 Level B. The packaged relays shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. The requirement for box closure, waterproofing and reinforcing shall be in accordance with method V of the PPP-B-636 appendix.

5.2.3 Level C. The level C packing for relays shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contact or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Relays, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a wood cap (storage aid 5) positioned over each load.

5.2.4.2 Level B. Relays, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that weather resistant fiberboard caps (storage aid 4) shall be used in lieu of wood caps.

5.2.4.3 Level C. Relays, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794 except that conformance to MIL-STD-147 is not required.

5.3 Marking. In addition to any special or other identification marking required by the contract (see 6.2), each unit, supplementary and exterior container and unitized load shall be marked in accordance with MIL-STD-129. The complete military or contractor's type or part number (including the FSCM), as applicable, shall be marked on all unit and intermediate packs in accordance with the identification marking provisions of MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

6. NOTES

6.1 Intended use. Relays conforming to this specification are intended for use in ac or dc ground training systems and similar controlled environment conditions to control the marking and breaking of circuits for electrically operated equipment and devices. Their principal areas of application are for training systems and controlled environment applications. This does not preclude the use of these relays in other military applications. This specification excludes thermal and completely solid state type relays. Consult MIL-STD-1346, Relays Selection and Application, as to selection and application.

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6.1.1 Contact rating. The contact ratings of relays covered by this specification are based on load endurance tests which establish the relay capability to switch rated loads, minimum loads from 100 mA up to rated loads, and low level loads. Performance may be unsatisfactory when contact current or voltage differs substantially from the rated values.

6.1.2 Voltage rating. Users of relays are cautioned that a 115/200 V rating implies that the relay is designed for use in a 3 phase system. The contacts may not be capable of switching 200 V. For circuit applications requiring relay contacts to interrupt 200 volts, the manufacturers of the relay should be consulted.

6.1.3 Intermediate current (minimum current). The current range within which the contact arc intensity is insufficient to consistently burn away carbonaceous materials on contact faces. The carbonaceous deposits are a result of breakdown of gases within a relay which originates from contaminants such as cleaning fluids, outgassing of materials within the relay, etc., which are broken down to carbon within the contact arc.

NOTE: Because intermediate current testing is exceedingly time consuming and because 50,000 operations of test cycles is sufficient to prove compliance capability, relays can be safely used for 100,000 operations applications.

6.1.4 Paralleling contacts. Contacts shall not be paralleled on the assumption that doing so will increase their switching capacity. Being paralleled for the sake of redundancy may result in make-before-break instead of break-before-make, due to one transfer contact operating slightly ahead of the other.

6.1.5 Latching relays. When latching relays are installed in equipment, the latch and reset coils should not be pulsed simultaneously. Coils should not be pulsed with less than the nominal coil voltage and the pulse width should be a minimum of three times the specified operate time of the relay. If these conditions are not followed, it is possible for the relay to be in the magnetically neutral position.

6.2 Ordering data.

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable specification sheet, and the part number.

c. Inspection of commercial packaging (see 4.6.3).

d. Levels of preservation and packing required (see 5.1 and 5.2).

e. Special or other identification marking if required (see 5.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List No. QPL-83520 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the qualified products list is the Electronic Support Directorate, Department of the Air Force; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), Dayton, OH 45444.

6.4 Intermetallic contact. The finishing of metallic areas to be placed in intimate contact by assembly presents a special problem, since intermetallic contact of dissimilar metals results in electrolytic couples which promote corrosion through galvanic action. To provide the required corrosion protection, intermetallic couples are restricted to those permitted by table VII. Table VII shows metals and alloys (or plates) by groups which have common electromotive forces (EMF) within 0.05 volt when coupled with a saturated calomel electrode in sea-water at room ambient

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TABLE VII. Compatible couples (see 6.4) 1/

Group No.	Metallurgical category	EMF (volt)	Anodic index (0.01 v)	Compatible couples
1	Gold, solid and plated; gold-platinum alloys; wrought platinum (most cathodic)	+ 0.15	0	
2	Rhodium plated on silver-plated copper	+ 0.05	10	
3	Silver, solid or plated; high silver alloys	0	15	
4	Nickel, solid or plated; monel metal, high nickel-copper alloys	- 0.15	30	
5	Copper, solid or plated; low brasses or bronzes; silver solder; German silver; high copper-nickel alloys; nickel-chromium alloys; austenitic corrosion-resistant steels	- 0.20	35	
6	Commercial yellow brasses and bronzes	- 0.25	40	
7	High brasses and bronzes; naval brass; Muntz metal	- 0.30	45	
8	18 percent chromium type corrosion-resistant steels	- 0.35	50	
9	Chromium, plated; tin, plated; 12 percent chromium type corrosion-resistant steels	- 0.45	60	
10	Tin-plate; terneplate; tin-lead solder	- 0.50	65	
11	Lead, solid or plated; high lead alloys	- 0.55	70	
12	Aluminum, wrought alloys of the duralumin type	- 0.60	75	
13	Iron, wrought, gray, or malleable; plain carbon and low alloy steels, armco iron	- 0.70	85	
14	Aluminum, wrought alloys other than duralumin type; aluminum, case alloys of the silicon type	- 0.75	90	
15	Aluminum, cast alloys other than silicon type; cadmium, plated and chromated	- 0.80	95	
16	Hot-dip-zinc plate; galvanized steel	- 1.05	120	
17	Zinc, wrought; zinc-base die-casting alloys; zinc, plated	- 1.10	125	
18	Magnesium and magnesium-base alloys, cast or wrought (most anodic)	- 1.60	175	

1/ Compatible couples - potential difference of 0.25 volt maximum between groups.

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temperatures. All members of a group are considered as completely compatible, one with the other. Compatible couples between groups have been specified in table VII based on a potential difference of 0.25 volt maximum. To simplify any arithmetic involved, table VII shown, in addition to EMF against a calomel electrode, a derived "anodic index" with group A (gold, etc.) as 0 and group 18 (magnesium, etc.) as 175. Subtraction of a lower group anodic index gives the EMF difference in hundredths of a volt.

6.4.1 Groups. Table VII sets up 18 primary groups. It may be noted that neither the metallurgical similarity or dissimilarity of metals is the parameter for selection of compatible couples. All members within a group, regardless of metallurgical similarity, are considered inherently nonsusceptible to galvanic action, when coupled with any member within the group; for example, such dissimilar metals as platinum and gold. Similarity, such basically dissimilar alloys as austenitic stainless steel, silver-solder, and low brass (all members of group 5) are inherently nonsusceptible when coupled together.

6.4.2 Compatibility graphs. Permissible couple series are shown in table XII by the graphs at the right. Members of groups connected by lines will form permissible couples. A O indicates the most cathode member of each series, a o an anodic member, and the arrow indicates the anodic direction.

6.4.3 Selection of compatible couples. Proper selection of metals in the design of equipment will result in fewer intermetallic contact problems. For example, for sheltered exposure, neither silver nor tin require protective finishes. However, since silver has an anodic index of 15 and tin 65, the EMF generated as a couple is 0.50 volt, which is not allowable by table VII. In this case, other metals or platings will be required. It should be noted that, in intermetallic couples, the member with the higher anodic index is anodic to the member with the lower anodic index and will be susceptible to corrosion in the presence of an electrolytic medium. If the surface area of the cathodic part is significantly greater than that of the anodic part, the corrosive attack on the contact area of the anodic part may be greatly itensified. Material selection for intermetallic contact parts, therefore, should establish the smaller part as the cathodic member of the couple, whenever practicable.

6.4.4 Plating. When base metals intended for intermetallic contact from couples not allowed by table VII, they are to be plated with those metals which will reduce the potential difference to that allowed by table VII.

6.5 Magnetic orientation. Magnetic orientation of a relay may be determined by energizing the coil with a known polarity and checking for attraction or repulsion of a permanent magnet to an external surface of the relay.

6.6 Glossary. The definitions listed below are not a complete glossary of relay terminology, but rather are intended as definitions of the technical terms as applied within the specification.

Actuator - That part of the relay that converts electrical power into force.

Armature - The moving magnetic member of an electro-magnetic relay structure.

Armature - The distance in air separating the magnetic portion of the armature, when fully restorted to its unoperated position, and the coil core face.

Backstop - That part of the relay which limits the movement of the armature away from the pole face or core. In some relays a normally closed contact may serve as backstop.

Bifilar winding - Two or more windings with the wire of each winding alongside the other, matching turn for turn; may be either inductive or noninductive.

Bistable (latching) relay - A two-position relay whose contacts transfer only as a result of coil energization (depending on design) (of either a particular polarity, or of a particular coil), remain in that position with no coil energization, and transfer to the alternate position only as a result of coil energization (of either the opposite polarity, or of the other coil, respectively).

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Bounce time - The interval between first make of the contact until the uncontrolled making and breaking of the contact ceases.

Break time - The break time is the amount of time that the movable contact is not closed to one stationary contact or the other, while the relay is being cycled, expressed as a percentage of the total cycle time. In telegraphy, the time that the contacts are open during transfer from mark to space and from space to mark. Usually expressed as a percent of total cycle time. Contact efficiency, the percentage of time that the contacts are conducting, is equal to 100 percent minus the break time percentage. All these times depend upon the coil current, supply voltage, and the wave form.

Chatter, contact - The undesired vibration of mating contacts resulting from uncompensated ac operation, or from external shock and vibration.

Coil - One or more windings on a common form.

Concentric winding - A coil with two or more insulated windings, wound radially one over the other.

Contacts - The current-carrying parts of a relay that open or close electrical circuits.

Contact, bifurcated - A forked or branched, contacting member so formed or arranged, as to provide dual contacting.

Contact, dry reed - A glass enclosed magnetically operated contact using reeds as the contracting members.

Contact, movable - A member of a contact arrangement that is moved directly by the actuating system.

Contact, stationary - A member of a contact arrangement that is not moved by the actuating system.

Contact arrangement - The combination of contact forms that make up the entire relay switching structure.

Contact bounce - Intermittent opening of contacts as they close, due to contact impact.

Contact force - The force exerted by a movable contact against a fixed contact when the contacts are closed.

Contact gap - The distance between a pair of mating relay contacts when the contacts are open.

Contact weld - A contact failure due to fusing of closed contacts, resulting in their failure to open.

Hermetically sealed relay - A relay contained within an enclosure that is sealed by fusion to insure a low rate of gas leakage. This usually refers to metal-to-metal, or metal-to-glass sealing.

Hold current or voltage - As the current or voltage applied to the coil on an operated relay is decreased, the value must be reached before any contact change occurs.

Intermediate current - The value of current at, or below, which there is insufficient current density at the mating contact surfaces to insure low contact resistance for the kind of contact material, shape, and pressure employed, or enough arcing to break down insulating deposits.

Latching relay - A bistable relay.

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Monostable relay - A (two-position and three position) relay whose contacts remain in or return to their normal (de-energized) position when no coil is energized.

Non-operating current or voltage - The maximum coil current or voltage at which a relay remains in the de-energized condition.

Operate time - The interval between the application of a step function input signal and closing of all normally open contacts. Bounce time is not included.

Pickup current or voltage - The current or voltage at which the armature seats against the coil core by assuming its fully operated position.

Relay - Most simply defined as an electrically controlled device that opens and closes electrical contacts to effect the operation of other devices in the same or another electrical circuit.

Relay, bistable - A two-position relay capable of remaining in either position when a coil signal is removed and of being driven to the opposite position only by applying a signal of suitable polarity. (Bistable relays are sometimes referred to as polar, latching relays which differ from latching relays having separate and non-polar inputs for latching and unlatching, respectively).

Relay, null-seeking - A relay of double-throw configuration and a stable center-off position. Either side of the double-throw contact can be activated by and for the duration of the polar actuating signal.

Relay, sensitive - A relay that operates on low input power, commonly defined as 100 milliwatts or less.

Release time - The interval between the trailing edge of a step function input signal and closing of all normally closed contacts. Bounce time is not included.

Saturation - The condition attained in a magnetic material when an increase in magnetizing (coil) current produces no appreciable increase in flux.

Sensitivity - Pickup current or voltage.

Talk, cross (cross-talk) - For the purpose of this specification, cross talk is defined as the signal in decibels (dB) down coupled across the open contacts of a switching device or the signal coupled between switching circuits within the same device.

Tandem winding - A coil having two or more windings, one behind the other, along the longitudinal axis.

Telephone-type relay - A relay with an end-mounted coil, an "L" shaped heelpiece, and contact springs parallel to the axis of the coil.

Time, operate - The time interval from coil energization to the functioning time of the last contact to function. Unless otherwise stated, it does not include contact bounce time.

Time, release - The time interval from coil de-energized to the functioning time of the last contact to function. Unless otherwise stated, it does not include bounce time.

Time, transfer - The time interval between the opening of the closed contact and the closing of the open contact and vice versa, of a break-make-contact combination.

Voltage, rated coil, or voltage nominal - The coil voltage at which the relay is intended to operate for the prescribed duty cycle.

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